

# The Paradox Docket: A Computable Classification of the Classical Paradoxes

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**Abstract.** For twenty-three centuries the word "paradox" has been handed out without an audit: it is worn by genuine logical dead ends, by harmless puzzles, and by plainly ill-posed questions. We conducted the audit. The classical collection — the liar and its family, Russell, Curry, Yablo, the crocodile, the Ship of Theseus and others — was run through a single executable instrument that, for every self-referential sentence, *counts the number of consistent classical solutions* and issues a passport: zero solutions — a genuine paradox (refusal forever); exactly one — the verdict is forced; two or more — not a paradox but a choice awaiting a decree. The result is a court docket: 21 case files and eight short verdicts, each reproducible with a single command. Few of the celebrities keep the title "paradox" once the papers are checked; most had been wearing it without documents. The taxonomy descends from Kripke and from Gupta-Belnap and is credited to them explicitly; what is new is executability: disputes that are decades or centuries old — is circularity vicious; is a loop hidden inside Yablo; is Curry just the liar; is the Ship of Theseus a paradox at all — receive measured rather than argued answers.

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## 1. Prologue: a word with no oversight

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Nobody polices the word "paradox" in philosophy. It is issued generously and for life: "the liar paradox", "the barber paradox", "the Ship of Theseus paradox", "the crocodile paradox" — one and the same word on the business cards of characters with entirely different fates. Yet the difference between them is not a nuance. Some questions have no consistent answer at all, and no future discovery will bring one. Others have exactly one answer — it merely arrives from an unexpected direction. Still others have several equally lawful answers, and the whole "abyss" reduces to the fact that nobody has taken responsibility for choosing. Wearing one word for all of them is like calling the plague, a runny nose, and nail-biting all "a disease".

This paper is an audit. We took the classical collection in its entirety, seated everyone at one table, and issued each a document. The instrument is executable: any verdict in this paper can be recomputed by the reader with one command on their own machine, or with one button in a browser (§7). By the last page every celebrity will hold a passport — and most will turn out to be impostors.

A word about genre, up front: we do not solve paradoxes and we do not propose a new theory of truth. We carry out a census with a measuring instrument — and discover that the census itself, when done by a machine, hands down verdicts in disputes that are decades and centuries old.

## 2. The court and its rules

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## 2.1. ZTL in one paragraph

The instrument is built on ZTL (Zero-Trust Logic), organized around a single principle: **truth is not granted on credit**. Every atomic claim has one of three statuses: *earned* — a completed verification has been produced (T); *refuted* — the verification came out against (F); *on credit* (Z) — no verification has been produced. Z is not "unknown" and not "half-true": it is an honest tag reading "unverified", and the logic forbids treating it as truth. Negation in ZTL is not a value-flipper but a verdict about the status of a verification; everything else follows from that asymmetry. This paragraph is all you need to read the paper. The full axiomatics, 394 machine-checked theorems (Lean 4, empty axiom lists) and the entire corpus of expeditions are in the canonical preprint: *ZTL — Zero-Trust Logic*, Zenodo, DOI 10.5281/zenodo.21318981 (concept DOI, resolving to the current version).

## 2.2. ZTLJudge: the judge

The first of the two working instruments is the judge (module `ztljudge`). Given a formula and a marking (what is earned, what refuted, what on credit), it returns three things: the **verdict** (true/false/refusal), the **grade** — the quality of the verdict (the highest grade, *hereditary*, means "earned under every possible future"; the lowest holds only until the first verification), and, most importantly, a **named list of weak links**: exactly which verifications are missing. The judge never says "something is wrong with your argument" — it says "your conclusion hangs on the unverified link j". In this paper the judge works in the eighth, acquitting verdict; the main work is done by the second instrument.

## 2.3. The passport office: the classifier

The second instrument is the passport office (module `zpassport`, expedition E18 of the ZTL corpus). Given a *system of definitions* — sentences referring to one another ("L asserts: L is false") — it does three things:

1. **It counts classical solutions.** It enumerates all consistent ways of assigning truth and falsehood to the sentences that agree with their own definitions. Then comes the arithmetic of fate: - **0 solutions → PARADOX**. The refusal is permanent: no decree and no discovery can help — every assignment contradicts the sentence's own definition. This is provable, not rhetorical. - **1 solution → INTRINSIC**. The sentence is ungrounded, but the verdict is *forced*: exactly one stipulation is possible. - **2+ solutions → UNDERDETERMINED**. Not a paradox but a blank form: any of the solutions can be legalized by decree, and the system closes up without breaking anything (this is a theorem — the "stipulation theorem"). - Off the loops there are two more passports: **INPUT** (an ordinary unverified input; refusal until verification) and **DOWNSTREAM** (the refusal is inherited from a culprit below, and the culprit is named).
2. **It takes the handwriting.** The convicted have a second identifying mark — the oscillation period: how fast the verdict "blinks" under recomputation (the liar — 2; Jourdain's carousel — 4). One passport, different signatures.
3. **It knows there are no further kinds.** Zero, one and "many" exhaust the possibilities of a finite system — the axis is provably complete, so any future "new paradox" of this genre has its place in the table reserved in advance.



*The Passport Office*

Fig. 1 — a self-referential sentence is counted, and issued its verdict

The control weight of the whole construction: a court that cannot convict is worth nothing. At this desk the liar receives precisely PARADOX — zero solutions, period 2, refusal forever. Every acquittal below is worth exactly as much as this conviction.

## 2.4. ZTLStudio: touch it yourself

Both instruments are available live: **ZTLStudio** (<https://ztl.vitalyreznik.com>) — a web interface over the very same core, no installation. The *Paradoxes* tab is the passport office: a drop-down list of classical examples, the formal record visible and editable, and a "Run on the core" button that returns the passport. The built-in AI translates from human language into the formal one — but it never issues verdicts; the studio warns explicitly: "*AI explanation unverified by definition — the verdict above is the authority.*" We return to the recipe in §7.

## 3. The docket: the whole collection at one table

Below is the main table. Every row is a measurement; the full reference run (`zclassify.py`, expedition E35) is attached as Appendix A and reproduces with a single command.

**Table 1. The court docket: 21 case files.**

| #  | Case                                       | One-line record                                                            | Passport                                 | Solutions | Period   | Negations |
|----|--------------------------------------------|----------------------------------------------------------------------------|------------------------------------------|-----------|----------|-----------|
| 1  | Liar                                       | $L \equiv \neg L$                                                          | <b>PARADOX</b>                           | 0         | 2        | 1         |
| 2  | Barber                                     | $\text{shaves}(b) \equiv \neg \text{shaves}(b)$                            | <b>PARADOX</b>                           | 0         | 2        | 1         |
| 3  | Grelling ("heterological")                 | $\text{het} \equiv \neg \text{het}$                                        | <b>PARADOX</b>                           | 0         | 2        | 1         |
| 4  | Russell's cell $R \in R$                   | $r \equiv \neg r$                                                          | <b>PARADOX</b>                           | 0         | 2        | 1         |
| 5  | Jourdain / crocodile                       | $R \equiv M, M \equiv \neg R$                                              | <b>PARADOX</b>                           | 0         | <b>4</b> | 1         |
| 6  | Odd 3-cycle                                | $a \equiv \neg b, b \equiv \neg c, c \equiv \neg a$                        | <b>PARADOX</b>                           | 0         | 2        | 3         |
| 7  | Curry with a real $\perp$                  | $\gamma \equiv (\gamma \rightarrow F)$                                     | <b>PARADOX</b>                           | 0         | 2        | —         |
| 8  | Truth-teller                               | $\tau \equiv \tau$                                                         | UNDERDETERMINED                          | 2         | 1        | 0         |
| 9  | Russell's twin $S \in S$                   | $s \equiv s$                                                               | UNDERDETERMINED                          | 2         | 1        | 0         |
| 10 | Optimistic crocodile (control)             | $R \equiv M, M \equiv R$                                                   | UNDERDETERMINED                          | 2         | 1        | 0         |
| 11 | Even 2-cycle                               | $A \equiv \neg B, B \equiv \neg A$                                         | UNDERDETERMINED                          | 2         | 2        | 2         |
| 12 | Even 4-cycle                               | (4 negations)                                                              | UNDERDETERMINED                          | 2         | 2        | 4         |
| 13 | Strong liar                                | $\sigma \equiv \neg \sigma \wedge \sigma$                                  | <b>INTRINSIC (<math>\sigma=F</math>)</b> | 1         | 1        | —         |
| 14 | Revenge / avenger                          | $\mu \equiv \neg(\mu \leftrightarrow \mu)$                                 | <b>INTRINSIC (<math>\mu=F</math>)</b>    | 1         | 1        | —         |
| 15 | Henkin-style                               | $h \equiv (h \rightarrow h)$                                               | <b>INTRINSIC (<math>h=T</math>)</b>      | 1         | 1        | —         |
| 16 | Yablo truncated at $n=3$                   | $s_i \equiv \bigwedge_{j \neq i} \neg s_j$                                 | GROUNDED                                 | —         | 1        | —         |
| 17 | Yablo truncated at $n=6$                   | —                                                                          | GROUNDED                                 | —         | 1        | —         |
| 18 | Theseus title contest                      | $\text{the}A \equiv \neg \text{the}B, \text{the}B \equiv \neg \text{the}A$ | UNDERDETERMINED                          | 2         | 2        | 2         |
| 19 | "Sameness with no criterion"               | $\text{same} \equiv \text{same}$                                           | UNDERDETERMINED                          | 2         | 1        | 0         |
| 20 | "The same person" (all observations match) | $S \equiv \text{obs} \wedge S, \text{obs}=T$                               | UNDERDETERMINED                          | 2         | 1        | —         |
| 21 | "The same person" (a mismatch)             | $S \equiv \text{obs} \wedge S, \text{obs}=F$                               | GROUNDED ( $S=F$ )                       | —         | 1        | —         |

Plus separate systems outside the table: Curry with a suspended  $\perp$  (case 7-bis, DOWNSTREAM — §4.5), Agrippa's dogma (UNDERDETERMINED + DOWNSTREAM with the culprit named), and the contingent liar in three worlds (§4.7).

Reminder cards, one sentence each: the **liar** — "this sentence is false"; the **barber** — shaves exactly those who do not shave themselves: does he shave himself?; **Grelling** — the word "heterological" means "not applying to itself": is it heterological?; **Russell** — the set of all sets that do not contain themselves: does it contain itself?; **Jourdain** — a postcard: the front reads "the sentence on the back is true", the back reads "the sentence on the front is false"; the **crocodile** — "I will return the child if you guess what I am about to do" — the mother: "you will not return him!"; **Curry** — "if this sentence is true, then  $\perp$ " (its everyday evil version: "if this sentence is true, then Santa Claus exists" — under classical treatment anything follows); the **truth-teller** — "this sentence is true"; **Yablo** — an infinite queue, each member saying

"everyone after me lies"; the **strong liar** — "I am lying, and I am right"; **Theseus** — every plank of the ship has been replaced, and a second ship was assembled from the old planks: which one is the real one?

## 4. Eight verdicts

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### 4.1. The liar is convicted — and has three twins

The liar is a genuine paradox: zero solutions, refusal forever, period 2. The confrontation is more interesting still. The barber, Grelling's "heterological" and Russell's famous cell gave *bit-for-bit identical testimony*: the same passport, the same zero solutions, the same period, the same parity. Four great names — **one paradox in four costumes**; "these are all the same thing" has been said for a century, and now it is a collated protocol.

Russell deserves a separate note: in the full universe (E11 of the ZTL corpus: the empty set, a self-containing set, and R) the catastrophe occupies **one cell** —  $R \in R$ . The other 8 of 9 membership facts are calmly grounded; Russell works perfectly well as a set for everyone except himself. Frege's whole system collapsed because classical logic demands an answer in every cell. A quarantine the size of one cell — versus demolishing the building.

### 4.2. The parity law

Take any circle of definitions and count the negations in it. **An odd count is a conviction; an even count is a blank form.** The truth-teller (zero negations) and the even cycles of length 2 and 4 all have exactly two lawful solutions, and the stipulation theorem guarantees: legalize either one and nothing breaks. The odd cycles (the liar, the triple) have zero solutions — refusal forever. The moral, worth pronouncing slowly: **it is not the circle that is vicious — it is the odd parity.** Half a century of "vicious circles" in the textbooks is, in fact, half a century of parity left uncounted.

### 4.3. Worsening the liar is useless

Intuition is sure that "I am lying AND I am right" is the liar squared. Measurement says the opposite: the strong liar has exactly **one** solution — it is false, forcedly, with no alternatives. No abyss: an ordinary false loudmouth, passport INTRINSIC. Its mirror — "if I am right, then I am right" — is forcedly *true*. Adding a contradiction does not deepen the pit — it fills it in: the more a sentence says about itself, the less freedom it has, and at "exactly one solution" the freedom runs out without ever reaching zero.

### 4.4. Yablo's alibi

Yablo's paradox is famous for the claim "a paradox without self-reference": an infinite queue of sentences, each saying "everyone after me lies". For half a century the dispute has run (Priest vs. Sorensen) over whether circularity is hidden inside it. Our contribution to the dispute is an alibi: **every finite truncation of the queue is fully grounded** (checked at  $n=3$  and  $n=6$  — not a single quarantined cell, every verdict computes). The paradoxicality does not live on any finite segment — all of it, without remainder, lives in the actual infinity. Whoever is prepared to pay for actual infinity pays for Yablo too; for whoever is not, Yablo is innocent for lack of a reachable corpus delicti.



## 4.5. Curry's two passports

Curry's sentence — "if this sentence is true, then  $\perp$ " — is the subject of an old dispute: is the implication to blame, the self-reference, or something third? Measurement splits the question into two cases. If  $\perp$  is a **real, grounded falsehood**, then Curry is literally the liar in an implication costume: zero solutions, period 2, the same passport. If instead  $\perp$  **itself hangs unverified** (defined over an ungrounded base), the passport changes: DOWNSTREAM — the refusal is not its own but inherited, and the culprit is named. The answer to the dispute: *it depends on what is fed to the arrow*. The two Currys now carry different documents, and confusing them is no longer obligatory.

## 4.6. The handwriting of the convicted

The liar blinks with period 2. Jourdain and the crocodile — the same article of the code, zero solutions — blink with period **4**: the verdict shift makes a round trip through both participants. The passport does not determine the handwriting — this is genuinely a second axis of classification (Gupta-Belnap revision signatures).

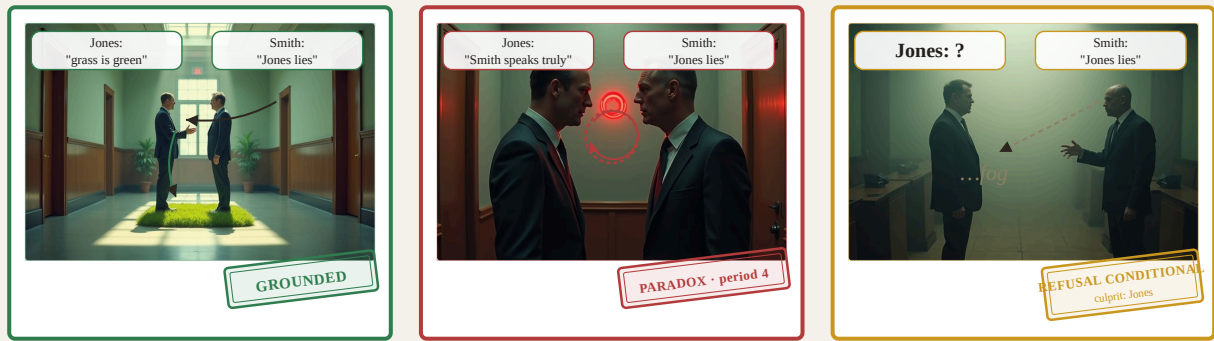
The crocodile also shows a second thing: how fragile the border between a conviction and a blank is — and who actually ends up holding the pen. Flip the mother's prediction to the optimistic one ("you WILL return him!") — one negation disappears, the parity clicks, and the conviction becomes a blank with two lawful solutions: he returned the child — the word was kept; he did not — the word was *also* kept. But notice who chooses between them: not the mother and not a court — **the crocodile, at his own pleasure**. The deal binds him in neither solution. And in the original, odd world the deal cannot be honored at all. The measurement (expedition E4) closes both worlds at once: the contract "I return  $\iff$  you guessed" **never earns truth** at the quarantined point — neither for the pessimist mother nor for the optimist. In legal translation: **the contract is void — no obligation ever arose**. For the pessimist it is void by impossibility of performance; for the optimist, by vacuity; in both worlds the mother decides nothing — her sentence is merely the content of someone else's loop. The famous "crocodile dilemma" is a dispute over a contract that was never concluded.

## 4.7. Paradoxicality is an accident

Kripke noticed what we have now measured. Smith says: "what Jones said is false." The sentence is one and the same — but its fate depends on the world:

- **World A.** Jones said "grass is green" (a truth). Smith's sentence is plain false. Everything grounded, case closed.
- **World B.** Jones was unlucky: he happened to say "Smith speaks truly." Two honest people, neither speaking about himself — and together they closed Jourdain's carousel: PARADOX, period 4, refusal forever.
- **World C.** Jones's words have not yet been verified. Passport: INPUT for Jones, DOWNSTREAM for Smith — the refusal is *conditional*, the culprit named; verify Jones and the case resolves either way.

**One sentence — three passports.** No filter over the text can catch paradoxes in advance: paradoxicality is not a property of a sentence but a concurrence of circumstances in the world the sentence talks about. A paradox is an event, not a text.



One and the same sentence of Smith's — three fates.

Fig. 2 — The Three Worlds of Smith and Jones

## 4.8. The acquitted

Our dilemma series (published as separate case studies) went through the same desk: the title contest of the Ship of Theseus, "sameness with no criterion", "is this the same person", Agrippa's dogma. All received the passports of **blanks**: two lawful solutions, a decree, nothing breaks; the dogma — DOWNSTREAM with the culprit named; "the same person", the moment one observation fails to match, grounds to "no" instantly. In the entire collection — **not one** paradoxical component. The court that convicted the liar (and Jourdain, and the odd cycles) acquitted Theseus — and that is what the acquittal is worth: the judge knows how to convict, so its "not a paradox" is a document, not politeness.

We foresee the metaphysician's objection: "planks, form, history — those are your three questions; I asked which ship is the same *in itself*." We do not leave this question unanswered — we issue it a document too. "Sameness in itself, free of any criterion" is the definition **same**  $:= \text{Tr}(\text{same})$ , which by its own construction rejects every producible criterion; the passport office hands it the truth-teller's blank: two lawful stipulations, neither of them earnable — the question itself has outlawed all evidence. The burden then flips: name one observation you would agree to count — and it will turn out to be one of the three criteria, with an instant answer; name none — and you have countersigned the passport yourself.

**Table 2. Certificates: old disputes — measured answers.**

| Dispute                             | Positions in the literature                                           | What the measurement showed                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Circularity is vicious"            | universal ban on self-reference (Tarski-style) vs. admission (Kripke) | parity is vicious, not the circle: odd — conviction, even — a blank (stipulation theorem)                                                                   |
| Yablo: is there a hidden loop?      | Priest (circular) vs. Sorensen (not)                                  | every finite segment grounded; all paradoxicality lives in the actual infinity                                                                              |
| Curry: who is to blame?             | implication vs. self-reference vs. detachment                         | depends on the status of $\perp$ : real falsehood $\rightarrow$ the liar; suspended $\rightarrow$ inherited refusal, culprit named                          |
| Russell: total ban or local repair? | type theory vs. restricted comprehension                              | the catastrophe = one cell of nine; types also ban the twin $S \in S$ , which is curable by decree                                                          |
| "The worsened liar is scarier"      | folklore intuition                                                    | the opposite: exactly one forced solution (INTRINSIC)                                                                                                       |
| The crocodile: what should he do?   | the sophists: a trap with no exit                                     | the contract is void — no obligation ever arose: unperformable for the pessimist (0 solutions), non-binding for the optimist (any action "honors" the deal) |
| Is Theseus a paradox?               | 2400 years of "yes" by default                                        | zero paradoxical components: facts plus a blank awaiting a decree                                                                                           |



The collection's group photo, after the document check

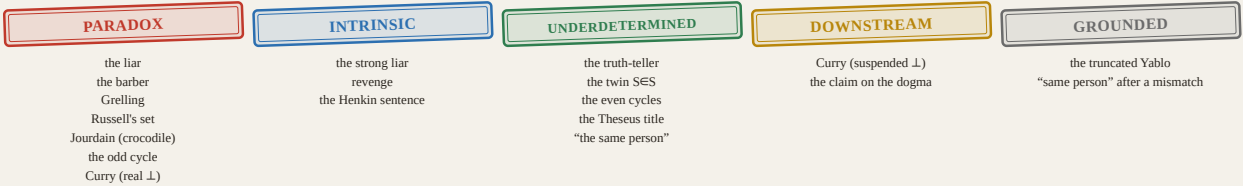


Fig. 3 — The Group Photo: who holds which passport



## 5. The limits of jurisdiction

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The passport office judges finite systems of definitions — the genre of self-reference. It honestly does **not** judge: the sorites (vagueness — there is no loop there, only a creeping boundary), the surprise exam (epistemic time), the lottery and St. Petersburg (probability), Berry and Richard (definability — that requires arithmetized naming, which a finite language does not express), Newcomb (decision theory). Some of these families are covered by other instruments of the ZTL corpus; some are covered by no one. We name the edges explicitly for a principled reason: a classification that does not know its own borders is just one more total theory. Knowing the edges is part of the strength, not a confession of weakness.

## 6. The honest frame

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The taxonomy's parents are named and not appropriated: the division into grounded / paradoxical / intrinsically forced is Kripke (1975); the oscillation signatures are the revision theory of Gupta and Belnap; the postcard itself is Jourdain (1913); the observation about contingency is Kripke's. The completeness of the axis (0/1/many exhaust a finite component) is an arithmetical fact, recorded in the ZTL corpus (E28).

What is ours is **executability and arbitration**: one instrument, one corpus, every row of the table a command, every verdict a run with a control weight. Throughout the paper "legitimate" means exactly one thing: *the version matched the measured passport*. It is a calibrated blessing, not an absolute one: we certify agreement with an instrument whose rules are open and themselves published.

## 7. Don't believe it — press the button

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Two ways to recompute this paper.

**In the browser (no installation):** ZTLStudio — <https://ztl.vitalyreznik.com>, the *Paradoxes* tab. The drop-down list holds this paper's entire collection: the liar, the barber, Grelling, Russell with his twin, Jourdain's postcard, both crocodiles, both Currys, the strong liar with its mirror, Yablo, the three worlds of the contingent liar, the Theseus title contest, Agrippa's dogma, "the same person" (variants such as Yablo-6 are one edited line away — hints are in the example descriptions). Pick an example, press *Validate* — the studio shows a human back-reading of the formal record — then *Run on the core*. The passport, the solution count, the stipulations — all on screen. The formal record is editable: flip one negation in a cycle and watch a conviction turn into a blank (§4.6).

**Locally:** the repository <https://github.com/inventor1975/ZTL>,

```
python3 zclassify.py
```

— the reference run of E35: the whole of Table 1 and all eight verdicts, each pinned by an assert; a failing assert is a refutation of this paper. The invitation to falsify comes with the genre.

## 8. Conclusion: a price list instead of a scare

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What has changed after the audit? Of the whole famous collection, few confirmed the title "paradox" once the documents were checked: the liar with his twins, the odd cycles, Jourdain with the crocodile, Curry under a real falsehood — and the infinite Yablo, if the reader is willing to pay for actual infinity. The rest are blanks awaiting a signature (the truth-teller, the even cycles, Theseus's title), forced verdicts (the strong liar and its family), and conditional refusals with named culprits. The word that scared students for twenty-three centuries has received a price list: every "defeat of reason" has its own price tag, and almost all the price tags turned out more modest than the signboard. The document check for one line takes milliseconds. The only expensive thing was the centuries lived without it.

## Acknowledgements and disclosure

The text, code and measurements were prepared in co-authorship with an AI (Claude Fable 5, Anthropic; Variant A — the human sets the tasks, verifies, and answers for the result). Curator — Vitaly Reznik. Instruments: ZTL (concept DOI 10.5281/zenodo.21318981), modules `zpassport` (E18) and `ztljudge`, the reference stand `zclassify.py` (E35), and the ZTLStudio web interface.

## References (sketch; to be finalized at typesetting)

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6. Curry, H. B. The Inconsistency of Certain Formal Logics. *JSL* 7 (1942).
7. Reznik, V. ZTL — Zero-Trust Logic. Zenodo, DOI 10.5281/zenodo.21318981 (v1.3, 2026).
8. (+ Russell, Grelling-Nelson, Prior/Geach as needed; completeness check at typesetting.)

## Appendix A. The reference run

The full output of `python3 zclassify.py`, identical to the one published in the repository at the publication commit:

| E35. THE DOCKET: MACHINE-CERTIFIED CLASSIFICATION OF PARADOXES |                                               |                 |             |   |   |
|----------------------------------------------------------------|-----------------------------------------------|-----------------|-------------|---|---|
| case                                                           |                                               | passport        | mod per par |   |   |
| liar                                                           | $L \equiv \neg L$                             | PARADOX         | 0           | 2 | 1 |
| barber                                                         | (alias)                                       | PARADOX         | 0           | 2 | 1 |
| Grelling                                                       | (alias)                                       | PARADOX         | 0           | 2 | 1 |
| Russell cell                                                   | $R \equiv R$ (alias)                          | PARADOX         | 0           | 2 | 1 |
| Jourdain/crocodile                                             | $R \equiv M, M \equiv \neg R$                 | PARADOX         | 0           | 4 | 1 |
| odd 3-cycle                                                    |                                               | PARADOX         | 0           | 2 | 3 |
| Curry, grounded                                                | $\perp: \gamma \equiv (\gamma \rightarrow F)$ | PARADOX         | 0           | 2 | - |
| truth-teller                                                   | $\tau \equiv \tau$                            | UNDERDETERMINED | 2           | 1 | 0 |
| Russell twin                                                   | $S \equiv S$                                  | UNDERDETERMINED | 2           | 1 | 0 |
| optimistic crocodile                                           | $R \equiv M, M \equiv R$                      | UNDERDETERMINED | 2           | 1 | 0 |
| even 2-cycle                                                   | $A \equiv \neg B, B \equiv \neg A$            | UNDERDETERMINED | 2           | 2 | 2 |
| even 4-cycle                                                   |                                               | UNDERDETERMINED | 2           | 2 | 4 |

|                                |                                          |                 |   |   |   |
|--------------------------------|------------------------------------------|-----------------|---|---|---|
| strong liar                    | $\sigma \equiv \neg\sigma\wedge\sigma$   | INTRINSIC       | 1 | 1 | - |
| revenge                        | $\mu \equiv \neg(\mu\leftrightarrow\mu)$ | INTRINSIC       | 1 | 1 | - |
| Henkin-style                   | $h \equiv (h\rightarrow h)$              | INTRINSIC       | 1 | 1 | - |
| Yablo trunc n=3                |                                          | GROUND          | - | 1 | - |
| Yablo trunc n=6                |                                          | GROUND          | - | 1 | - |
| Theseus title contest          |                                          | UNDERDETERMINED | 2 | 2 | 2 |
| Theseus 'same', criterion-free |                                          | UNDERDETERMINED | 2 | 1 | 0 |
| person corecursion, obs=T      |                                          | UNDERDETERMINED | 2 | 1 | - |
| person corecursion, obs=F      |                                          | GROUND          | - | 1 | - |

### V1. Parity law: no exception on pure negation cycles  
ok odd (1,3)  $\Rightarrow$  PARADOX/0 models; even (0,2,4)  $\Rightarrow$  UNDERDETERMINED/2, every even stipulation grounds cleanly – Kripke transported, total

### V2. Alias certificates: one paradox, four costumes  
ok barber = Grelling = RER = liar (passport, models, period)  
ok full Russell universe: E11 – one cell quarantined, 8/9 grounded; minimal surgery certified; type theory also bans the curable twin

### V3. Yablo: no finite stage is paradoxical  
ok n=3, n=6 GROUND outright – paradoxicality lives only in the actual infinity (measured side in the Priest–Sorensen dispute)

### V4. The intrinsic trio: forced verdicts on both sides  
ok strong liar forced  $\sigma=F$ ; revenge forced  $\mu=F$ ; Henkin forced  $h=T$  – 'worse' self-reference is TAMER: one model, stipulation forced

### V5. Two Currys, two passports –  $\perp$  decides  
ok grounded  $\perp \Rightarrow$  PARADOX (Curry IS the liar in arrow costume)  
ok suspended  $\perp \Rightarrow$  DOWNSTREAM (culprits [' $\perp$ '], refusal conditional)

### V6. Period: the second axis is real  
ok liar 2 vs carousel 4 under one passport; one flipped negation moves the carousel across the parity line

### V7. The dilemma series files into the docket  
ok Theseus contest / criterion-free same / person / Agrippa's dogma: decree-resolvable non-paradoxes, same instrument, same table

### V8. The contingent liar: paradoxicality is empirical  
ok world A (Jones told a truth): GROUND – S is ordinary falsehood  
ok world B (Jones said 'Smith speaks truly'): PARADOX, period 4 – the unlucky configuration IS the Jourdain carousel  
ok world C (Jones unverified): INPUT + DOWNSTREAM (culprits ['J'], refusal conditional)  
ok one sentence, three passports – a paradox is an EVENT, not a text; no syntactic sieve can quarantine paradoxes in advance

E35: docket complete – every row pinned, every verdict certified.  
Genre borders stand outside the axis by design: tortoise & sensor  $\rightarrow$  warranty genre; sorites, surprise exam, lottery, Berry  $\rightarrow$  other instruments. The liar earns its title; the impostors are named.

## Appendix B. A pocket glossary

- **Earned (T)** — a completed verification has been produced.
- **On credit (Z)** — no verification exists; the logic forbids spending such "truth".
- **Quarantine** — the status of a sentence to which lazily-grounded recomputation assigns no classical value; the passport explains why.
- **Passport** — the type of refusal: PARADOX / INTRINSIC / UNDERDETERMINED / INPUT / DOWNSTREAM.
- **Stipulation (decree)** — an external decision legalizing one of the admissible values; the stipulation theorem guarantees safety for UNDERDETERMINED and impossibility for PARADOX.
- **Period** — the blinking frequency of the verdict under recomputation; the convict's handwriting.